School/Faculty:	**Computing/Engineering**	**Page:**	2 of 6
Program name:	**Bachelor of Computer Science (Software Engineering)**		
Course code:	**SECJ 4383**	**Academic Session/Semester:**	**2022/23/1**
Course name:	**Software Construction**	**Pre/co requisite (course name and code, if applicable):**	**Software Engineering**
Credit hours:	**3**		

COURSE INFORMATION

School/Faculty:	Computing/Engineering	**Page:**	1 of 6
Program name:	Bachelor of Computer Science (Software Engineering)		
Course code:	SECJ 4383	**Academic Session/Semester:**	2023/24/1
Course name:	Software Construction	**Pre/co requisite (course name and code, if applicable):**	Software Engineering
Credit hours:	3		

Course synopsis	This course equips the students with knowledge in software construction particularly on how software engineering approaches assist software development. The knowledge unit for thiscourse area emphasize the following topics; design foundation, reuse, design by contract, components and advance components, framework, refactoring and code smells. The objective of this course is to introduce students with key software construction practices in software development and give practical experience to the students in developing a software using appropriate software construction techniques and tools.			
Course coordinator (if applicable)	Dr. Adila Firdaus Binti Arbain			
Course lecturer(s)	**Name**	**Office**	**Contact no.**	**E-mail**
	Dr. Adila Firdaus Binti Arbain	N28A-201- 02		adilafirdaus@utm.my
	Dr. Mohd Adham Bin Isa			mohdadham@utm.my
	PM Dr Hishammuddin Bin Asmuni @ Hasmuni			hishamudin@utm.my

Mapping of the Course Learning Outcomes (CLO) to the Programme Learning Outcomes (PLO), Teaching & Learning (T&L) methods and Assessment methods:

School/Faculty:	**Computing/Engineering**	**Page:**	2 of 6
Program name:	**Bachelor of Computer Science (Software Engineering)**		
Course code:	**SECJ 4383**	**Academic Session/Semester:**	**2022/23/1**
Course name:	**Software Construction**	**Pre/co requisite (course name and code, if applicable):**	**Software Engineering**
Credit hours:	**3**		

No.	**CLO**	**PLO (Code)**	***Taxonomies and **generic skills**	**T&L methods**	****Assessment methods**
CLO 1	Apply concepts of software construction in a software development.	PLO1 (KW)	C4	Lecture, Tutorial, active learning	Q, PS, T
CLO 2	Apply appropriate software engineering methods and tools in constructing a software.	PLO2 (TE)	C4	Lab work, Tutorial	A,PS, F
CLO 3	Analyse the quality of a software construction in a software development.	PLO3 (PS)	C6, A3	Lecture, Tutorial, Project- based learning	GP, F
=CLO 4	Work in a team to apply software construction knowledge in developing a medium application.	PLO8 (AD)	AD3, AD4	Project- based learning	A, GP

Refer *Taxonomies of Learning and **UTM's Graduate Attributes, where applicable for measurement of outcomes achievement
***T T – Test; Q – Quiz; HW – Homework; L – Lab, GR – Group Project; PR – Personal Report; F – Final Exam etc.

Prepared by Name: ' Mohd Adham Isa Signature: Date:	Certified By Name: Radziah Mohamad Signature: Date:

Details on Innovative T&L practices:

No.	**Type**	**Implementation**
1.	Active learning	Conducted through in-class activities
2.	Lab works & Tutorial	Conducted through in-class
3.	Project-based learning	Conducted through design assignments. Students in a group of 4 are given two assignments task and one projects related to the software construction involving the knowledge to adapt and apply known software engineering.

School/Faculty:	**Computing/Engineering**	**Page:**	2 of 6
Program name:	**Bachelor of Computer Science (Software Engineering)**		
Course code:	**SECJ 4383**	**Academic Session/Semester:**	**2023/24/1**
Course name:	**Software Construction**	**Pre/co requisite (course name and code, if applicable):**	**Software Engineering**
Credit hours:	**3**		

Weekly Schedule: The test case selection is the first challenge concerning the efficiency of multi-objective test case selection.

Week 1 19/3-23/3 23/3 (Thursday)- Sultan Johor's -First Day Ramadan	**Introduction to Software Construction** Programming paradigm, OO Concepts, Introduction to Git
Week 2 26/3-30/3 26/3 (Johor Holiday)	**Design Foundation** Design principles, Properties of design, Design guideline, Code management
Week 3 2/4-6/4	**Software Reuse** Definition and taxonomy, Reusability, Process and technology, Cost and benefit Problem Solving 1

Week 4 9/4-13/4	**Design by Contract** Violating a contract, DbC in programming language, Contract and Inheritance, Defensive programming Problem Solving 2
Week 5 16/4-20/4	**DeveOps** Assignment 1: Due date Week 7

Week 6 23/4-27/4	**SEMESTER BREAK**
Week 7 30/4-4/5 1/5(Monday) Labour Day 4/5(Thursday) -Wesak Day	DevOps part II
Week 8 7/5-11/5	Mid-Term THE 9/5 (Tuesday)- 8.00 pm to 10.00 pm
Week 9 14/5-18/5	**Domain Modelling** Application domain, Identification and modelling of domain concepts, Discovering classes Problem Solving 3
Week 10 21/5-25/5 Week 11 28/5-1/6	**Components and Advanced Component Concept** Properties of components, Components model, OSGi, Managed components (Spring framework), Components View Project 1 : Due Date; Week 15 Assignment 2: Due date Week 15
Week 12 4/6-8/6 5/6 (Monday) Agong's Birthday	**Framework** Class libraries, Framework classification, Plug-extension mechanism, Hot-spot driven development, Framework Instantiation
Week 13 11/6-15/6	**Refactoring** Re-engineering, Refactoring types, Process and guidelines Problem Solving 4
Week 14 18/6-22/6	**Code Smells** Code and design smells, Detecting bad smells
Week 15 25/6-29/6 28/6 (Wednesday) Aidil Adha EID	**Project Presentation**

Transferable skills (generic skills learned in course of study which can be useful and utilised in other settings):

~~Able to adapt knowledge of software engineering in different environment technology setting.~~

Student learning time (SLT) details:

Distribution of student Learning Time (SLT) Course content outline					Teaching and Learning Activities		TOTAL SLT
	Guided Learning (Face to Face)				Guided Learning Non-Face to Face	Independent Learning Non-Face to face	
CLO	**L**	**T**	**P**	**O**			
CLO 1	10h	5h				14.4h	**29.4h**
CLO 2	4h		8h		11.2h	11.2h	**34.4h**
CLO 3	3h	3h				17.2h	**23.2h**
CLO 4	3h			1h	9h		**13h**
Total SLT	**20h**	**8h**	**8h**	**1h**	**21.4h**	**41.6h**	**100h**

Continuous Assessment		PLO	Percentage	Total SLT
1	Problem Solving 1 - 4	KW, TE	10	5h
2	Test	KW	20	2h
3	Assignment 1	TE	8	3h
4	Assignment 2	KW, AD	7	4h
5	Project development	AD,PS	20	3h
Final Assessment			Percentage	Total SLT
1	Final Examination, Section A	TE	20	1.5
2	Final Examination, Section B	PS	15	**1.5**
Grand Total			**100**	**120h**

L: Lecture, T: Tutorial, P: Practical, O: Others

Special requirement to deliver the course (e.g: software, nursery, computer lab, simulation room):

Git

Trello

Eclipse/ NetBeans

Learning resources:

Text book (if applicable)

Note: This course should refer mainly to research papers like published journals and conference proceedings that describe current issues and solutions in software development in general, and specifically in each phase of software development.

Main references

1. McConnell s. (2004). "Code Complete", Microsoft Press

Additional references

Online

Academic honesty and plagiarism: *(Below is just a sample)*

Assignments are individual tasks and NOT group activities (UNLESS EXPLICITLY INDICATED AS GROUP ACTIVITIES) Copying of work (texts, simulation results etc.) from other students/groups or from other sources is not allowed. Brief quotations are allowed and then only if indicated as such. Existing texts should be reformulated with your own words used to explain what you have read. It is not acceptable to retype existing texts and just acknowledge the source as a reference. Be warned: students who submit copied work will obtain a mark of **zero** for the assignment and disciplinary steps may be taken by the Faculty. It is also unacceptable to do somebody else's work, to lend your work to them or to make your work available to them to copy.

Other additional information (Course policy, any specific instruction etc.):

1. Attendance is compulsory and will be taken in every lecture session. Student with less than 80% of total attendance is not allowed to sit for final exam.
2. Students are required to behave and follow the University's dressing regulation and etiquette all the time.
3. Exercises and tutorial will be given in class and some may be taken for assessment. Students who do not do the exercise will lose the coursework marks for the exercise.
4. Assignments must be submitted on the due dates. Some points will be deducted for late submissions. Assignments submitted three days after the due date will not beaccepted.
5. Make up exam will not be given, except to students who are sick and submit medical certificate confirmed by UTM panel doctors. Make up exam can only be given within one week of the initial date of exam.
6. The following table shows the mapping of the course assessments with the CLO andPLO.

No	Assessment	PLO 1	PLO 2	PLO 3	PLO 8	TOTAL
		CLO 1	CLO 2	CLO 3	CLO 4	
1	Assignment 1	4			3	7
2	Assignment 2		5		3	8
3	Problem Solving	5	5			10
4	Test	10	10			20
5	Project			15	5	20
6	Final Exam		20	15		35
TOTAL PLO		19	40	30	11	100

Disclaimer:

All teaching and learning materials associated with this course are for personal use only. The materials are intended for educational purposes only. Reproduction of the materials in any form for any purposes other than what it is intended for is prohibited.
While every effort has been made to ensure the accuracy of the information supplied herein, Universiti Teknologi Malaysia cannot be held responsible for any errors or omissions.